Developer API Test Cases (Postman)

AgriOpportunity API v1 · Collection: postman/AgriOpportunity.postman_collection.json · Environment: postman/AgriOpportunity.postman_environment.json

1. Summary

Item	Result
Requests executed	51 (failed: 0)
Assertions	150 (failed: 0)
Test cases passed	51 / 51
Runner	Newman (Postman CLI) · run in order, one iteration
Data	Freshly seeded database (SEED_ON_START=true)
Automated integration tests	backend/tests/api.test.ts — 42 Vitest + Supertest tests, all passing

2. How to run

Postman app: Import both JSON files, choose the *AgriOpportunity - Render* environment, set **baseUrl** to the deployed API, then Collections → AgriOpportunity → Run. Keep the default order. **Terminal:** `npx newman run postman/AgriOpportunity.postman_collection.json -e postman/AgriOpportunity.postman_environment.json --env-var baseUrl=https://<your-service>.onrender.com`. Each run registers a new youth with a timestamped email and deletes it at the end, so the collection can be re-run against the same database. Every request also asserts a response time under 3 seconds (warm server).

3. Golden journey coverage

Step	Test cases
Set profile	TC-09 register, TC-13 matches blocked without profile, TC-14 invalid profile, TC-15 set profile
See matches	TC-17 match count + ranked list + 3 reasons, TC-18 partial match explained, TC-19 only open items
Save one	TC-27 save with 3-day alert, TC-29 saved list (offline source)
Receive a closing-date alert	TC-30 alert due with message, TC-31 calendar file with reminder, TC-32/33 dismiss

4. Test cases

ID	Test case	Method & endpoint	Auth	Input / precondition	Expected result	Actual	Status
Meta							
TC-01	**Health check** API and database reachable	`GET /health`	None	—	200, status ok, database up	200 · 28 ms · 3 checks	**PASS**
TC-02	**Reference lists** Provinces, fields, levels and types in one call	`GET /meta`	None	—	200, 9 provinces, 7 fields, 6 levels, types	200 · 4 ms · 3 checks	**PASS**
TC-03	**Unknown route** Consistent error shape	`GET /nope`	None	—	404 NOT_FOUND	404 · 2 ms · 3 checks	**PASS**
Auth							
TC-04	**Login as admin** Seeded admin signs in	`POST /auth/login`	None	email=admin@agriopportunity.test, password=Admin@1234	200, role admin	200 · 84 ms · 3 checks	**PASS**
TC-05	**Login as youth with profile** Login returns the matching profile	`POST /auth/login`	None	email=lerato@agriopportunity.test, password=Youth@1234	200, profile present, no hash	200 · 81 ms · 3 checks	**PASS**
TC-06	**Wrong password** Generic message	`POST /auth/login`	None	email=lerato@agriopportunity.test, password=wrong-pass1	401 generic message	401 · 81 ms · 3 checks	**PASS**
TC-07	**Register without consent** POPIA consent required	`POST /auth/register`	None	name=No Consent, email=nc{{runId}}@agriopportunity.test, password=Passw0rd!	400 consent error	400 · 4 ms · 3 checks	**PASS**
TC-08	**Register with weak password** Password policy	`POST /auth/register`	None	name=Weak, email=weak{{runId}}@agriopportunity.test, password=abc, consent=true	400 password error	400 · 17 ms · 3 checks	**PASS**
TC-09	**Register new youth** Golden journey start: a new youth account	`POST /auth/register`	None	name=Test Youth, email=youth{{runId}}@agriopportunity.test, password=Youth@1234, consent=true	201, role youth, no profile	201 · 86 ms · 3 checks	**PASS**
TC-10	**Duplicate email** Email unique	`POST /auth/register`	None	name=Dup, email=youth{{runId}}@agriopportunity.test, password=Youth@1234, consent=true	409	409 · 4 ms · 3 checks	**PASS**
TC-11	**Me without token** Protected route	`GET /auth/me`	None	—	401	401 · 2 ms · 3 checks	**PASS**
TC-12	**Tampered token** JWT signature checked	`GET /auth/me`	Tampered token	—	401	401 · 2 ms · 3 checks	**PASS**
Profile & matches							
TC-13	**Matches before profile** Ask for a profile first	`GET /opportunities/matches`	New youth	—	409 set up profile	409 · 4 ms · 3 checks	**PASS**
TC-14	**Invalid profile** Validation per field	`PUT /profile`	New youth	field=astronomy, province=Mars, level=9	400 field, province, level errors	400 · 4 ms · 3 checks	**PASS**

ID	Test case	Method & endpoint	Auth	Input / precondition	Expected result	Actual	Status
TC-15	**Set profile** Golden journey step 1: set profile	`PUT /profile`	New youth	field=crop-production, province=Limpopo, level=3	200, profile saved	200 · 5 ms · 3 checks	**PASS**
TC-16	**Admin cannot set a profile** Role check	`PUT /profile`	Admin	field=crop-production, province=Limpopo, level=3	403	403 · 2 ms · 3 checks	**PASS**
TC-17	**Match count and ranked list** Golden journey step 2: "N opportunities match your profile", ranked	`GET /opportunities/matches`	New youth	—	200, count >= 3, score desc, soonest first, 3 reasons	200 · 9 ms · 3 checks	**PASS**
TC-18	**Partial match explains the gap** Explainable matching	`GET /opportunities/{{partialId}}`	New youth	—	200, failed check says Needs... or Only in...	200 · 5 ms · 3 checks	**PASS**
TC-19	**Expired items never match** Only open opportunities	`GET /opportunities/matches`	New youth	—	200, all isOpen	200 · 8 ms · 3 checks	**PASS**
Search							
TC-20	**Guest search with province** Province filter includes national items	`GET /opportunities?province=Limpopo`	None	—	200, Limpopo or National only, no match info	200 · 4 ms · 3 checks	**PASS**
TC-21	**Type + closing within 7 days** Closing-soon filter	`GET /opportunities?type=internship&closingWithin=7`	None	—	200, internships closing in 7 days	200 · 4 ms · 3 checks	**PASS**
TC-22	**Default sort is soonest closing** Sort by closing date	`GET /opportunities?limit=50`	None	—	200, ascending closing dates	200 · 5 ms · 3 checks	**PASS**
TC-23	**Signed-in search includes match** Personalised search	`GET /opportunities?limit=3`	Youth (Lerato)	—	200, match + saved fields	200 · 5 ms · 3 checks	**PASS**
TC-24	**SQL injection attempt** Parameterised SQL	`GET /opportunities?q=%27%3B%20DROP%20TABLE%20opportunities%3B%20--`	None	—	200, empty, table intact	200 · 3 ms · 3 checks	**PASS**
TC-25	**Invalid filter** Query validation	`GET /opportunities?type=party`	None	—	400	400 · 1 ms · 3 checks	**PASS**
TC-26	**Invalid id** UUID check	`GET /opportunities/not-a-uuid`	None	—	400	400 · 2 ms · 3 checks	**PASS**
Save & alerts							
TC-27	**Save top match with 3-day alert** Golden journey step 3: save one	`POST /opportunities/{{matchId}}/save`	New youth	daysBefore=3	201, saved, alert 3 days	201 · 8 ms · 3 checks	**PASS**
TC-28	**Invalid reminder** 0-30 days only	`POST /opportunities/{{matchId}}/save`	New youth	daysBefore=90	400	400 · 2 ms · 3 checks	**PASS**
TC-29	**Saved list** Offline saved list source	`GET /me/saved`	New youth	—	200, contains saved item	200 · 4 ms · 3 checks	**PASS**

ID	Test case	Method & endpoint	Auth	Input / precondition	Expected result	Actual	Status
TC-30	**Closing-date alert is due** Golden journey step 4: receive a closing-date alert (item closes in 2 days)	`GET /me/alerts`	New youth	—	200, alert in due with message	200 · 3 ms · 3 checks	**PASS**
TC-31	**Calendar file** Phone calendar reminder	`GET /opportunities/{{matchId}}/calendar.ics?daysBefore=3`	None	—	200, text/calendar with VALARM -P3D	200 · 2 ms · 3 checks	**PASS**
TC-32	**Dismiss alert** Alert handled	`PATCH /me/alerts/{{alertId}}/dismiss`	New youth	—	204	204 · 3 ms · 2 checks	**PASS**
TC-33	**Dismissed alert gone** Dismiss works	`GET /me/alerts`	New youth	—	200, not in due	200 · 2 ms · 3 checks	**PASS**
Apply							
TC-34	**Apply with form** Simple form application	`POST /opportunities/{{matchId}}/apply`	New youth	motivation=I studied a diploma in plant production and farm vegetables with my family.	201 submitted	201 · 5 ms · 3 checks	**PASS**
TC-35	**Apply twice** One application each	`POST /opportunities/{{matchId}}/apply`	New youth	motivation=Trying to apply a second time for this.	409	409 · 5 ms · 3 checks	**PASS**
TC-36	**Link opportunity refuses form** Send people to the original application	`POST /opportunities/{{linkId}}/apply`	New youth	motivation=I would like to apply for this one please.	400 apply on website	400 · 3 ms · 3 checks	**PASS**
TC-37	**Motivation too short** Validation	`POST /opportunities/{{partialId}}/apply`	New youth	motivation=Hi	400	400 · 2 ms · 3 checks	**PASS**
TC-38	**My applications** Track applications	`GET /me/applications`	New youth	—	200, 1 item	200 · 3 ms · 3 checks	**PASS**
Admin							
TC-39	**Youth blocked from admin** Admin only	`GET /admin/opportunities`	Youth (Lerato)	—	403	403 · 2 ms · 3 checks	**PASS**
TC-40	**Past date and missing link rejected** Business rules	`POST /admin/opportunities`	Admin	title=Bad opportunity, type=job, organisation=Test Org, description=A description that is long enough to pass., field=any, province=Natio...	400 closingDate + applyUrl	400 · 3 ms · 3 checks	**PASS**
TC-41	**Create opportunity** Admin creates	`POST /admin/opportunities`	Admin	title=Seedling Nursery Internship {{runId}}, type=internship, organisation=Test Nursery (demo), description=Six months in a seedling nurs...	201 open	201 · 5 ms · 3 checks	**PASS**
TC-42	**New item appears in matches** Matching picks it up	`GET /opportunities/matches`	Youth (Lerato)	—	200, includes new item	200 · 8 ms · 3 checks	**PASS**
TC-43	**Edit opportunity** Admin edits	`PATCH /admin/opportunities/{{createdId}}`	Admin	stipend=R5 000 per month	200 updated	200 · 7 ms · 3 checks	**PASS**

ID	Test case	Method & endpoint	Auth	Input / precondition	Expected result	Actual	Status
TC-44	**Expire opportunity** Admin expires	`PATCH /admin/opportunities/{{createdId}}/expire`	Admin	—	200, closed	200 · 4 ms · 3 checks	**PASS**
TC-45	**Expired hidden from search** Expiry works	`GET /opportunities?q=Seedling%20Nursery%20Internship%20{{runId}}`	None	—	200, empty	200 · 3 ms · 3 checks	**PASS**
TC-46	**Cannot save expired** Closed is closed	`POST /opportunities/{{createdId}}/save`	Youth (Lerato)	—	409	409 · 3 ms · 3 checks	**PASS**
TC-47	**Applications for opportunity** Admin sees applicants	`GET /admin/opportunities/{{matchId}}/applications`	Admin	—	200, >= 1	200 · 3 ms · 3 checks	**PASS**
TC-48	**Stats** Dashboard numbers	`GET /admin/stats`	Admin	—	200 counters	200 · 4 ms · 3 checks	**PASS**
Privacy							
TC-49	**Unsave** Remove saved + alert	`DELETE /opportunities/{{matchId}}/save`	New youth	—	204	204 · 4 ms · 2 checks	**PASS**
TC-50	**Delete my account** POPIA: right to deletion	`DELETE /auth/me`	New youth	—	204	204 · 3 ms · 2 checks	**PASS**
TC-51	**Deleted account cannot log in** Data gone	`POST /auth/login`	None	email=youth{{runId}}@agriopportunity.test, password=Youth@1234	401	401 · 3 ms · 3 checks	**PASS**

5. Automated integration tests (Vitest + Supertest)

`cd backend && DATABASE_URL=postgres://.../agriopportunity_test npm test` runs 42 tests against a real PostgreSQL database: health and reference lists; auth (consent, duplicate email, generic login error, profile in /me, tampered token); the golden journey (matches need a profile, invalid profile, set profile, match count 3 with ranked and explained list, partial match reason, save with alert, re-save updates alert, saved list, alert due, dismiss, calendar file, unsave); search (province includes national, type and closing-soon filters, sort, personalised match info, SQL-injection safety, bad filters and ids); apply (form, duplicate, link refuses form, my applications); admin (role check, business rules, create appears in matches, edit, edit keeps rules, expire hides it, cannot save expired, applications and stats); starting data (a learnership and a job in every province, no duplicates on restart); matching rules unit test; privacy (account deletion). Result: **42 passed, 0 failed**.